

18.1 FEEDBACK CONCEPTS

A typical feedback connection is shown in Fig. 18.1. The input signal, V_s , is applied to a mixer network, where it is combined with a feedback signal, V_f . The difference of these signals, V_i , is then the input voltage to the amplifier. A portion of the amplifier output, V_o , is connected to the feedback network (β), which provides a reduced portion of the output as feedback signal to the input mixer network.

If the feedback signal is of opposite polarity to the input signal, as shown in Fig. 18.1, negative feedback results. While negative feedback results in reduced overall voltage gain, a number of improvements are obtained, among them being:

1. Higher input impedance.
2. Better stabilized voltage gain.
3. Improved frequency response.
4. Lower output impedance.
5. Reduced noise.
6. More linear operation.

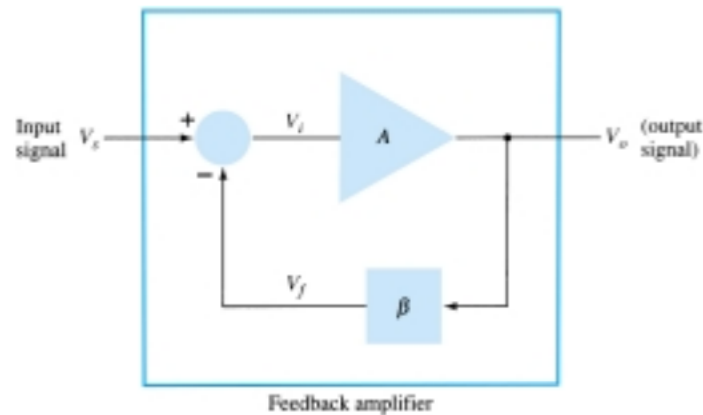


Figure 18.1 Simple block diagram of feedback amplifier.

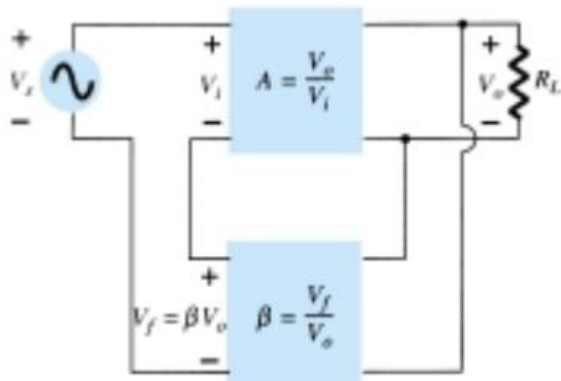
18.2 FEEDBACK CONNECTION TYPES

There are four basic ways of connecting the feedback signal. Both *voltage* and *current* can be fed back to the input either in *series* or *parallel*. Specifically, there can be:

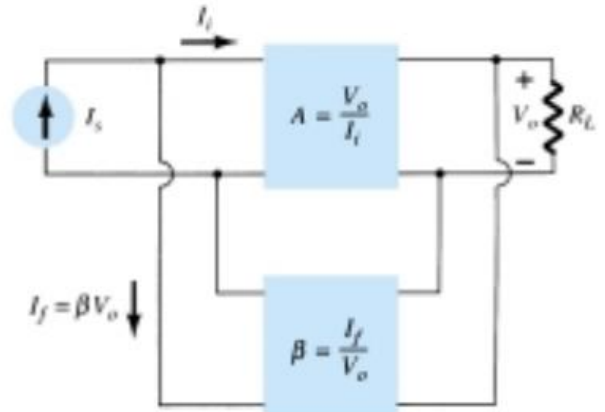
1. Voltage-series feedback (Fig. 18.2a).
2. Voltage-shunt feedback (Fig. 18.2b).
3. Current-series feedback (Fig. 18.2c).
4. Current-shunt feedback (Fig. 18.2d).

In the list above, *voltage* refers to connecting the output voltage as input to the feedback network; *current* refers to tapping off some output current through the feedback network. *Series* refers to connecting the feedback signal in series with the input signal voltage; *shunt* refers to connecting the feedback signal in shunt (parallel) with an input current source.

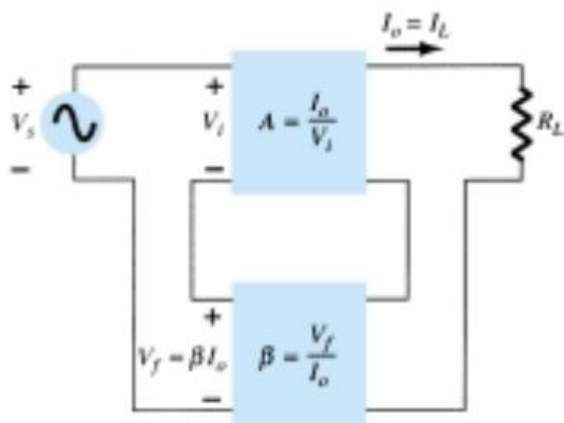
Series feedback connections tend to *increase* the input resistance, while shunt feedback connections tend to *decrease* the input resistance. Voltage feedback tends to *decrease* the output impedance, while current feedback tends to *increase* the output impedance. Typically, higher input and lower output impedances are desired for most



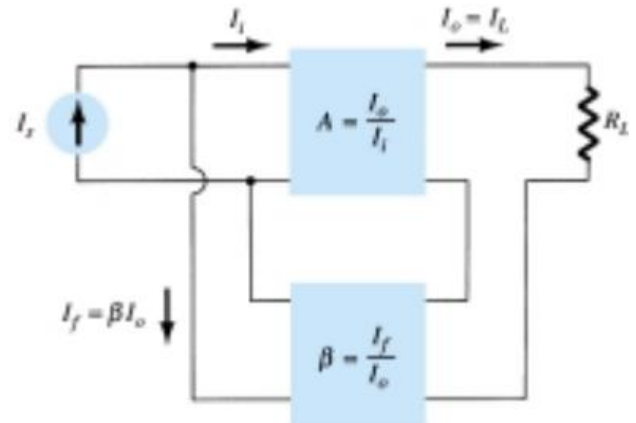
(a)



(b)



(c)



(d)

Figure 18.2 Feedback amplifier types: (a) voltage-series feedback, $A_f = V_o/V_s$; (b) voltage-shunt feedback, $A_f = V_o/I_s$; (c) current-series feedback, $A_f = I_o/V_s$; (d) current-shunt feedback, $A_f = I_o/I_s$.

cascade amplifiers. Both of these are provided using the voltage-series feedback connection. We shall therefore concentrate first on this amplifier connection.

Gain with Feedback

In this section we examine the gain of each of the feedback circuit connections of Fig. 18.2. The gain without feedback, A , is that of the amplifier stage. With feedback, β , the overall gain of the circuit is reduced by a factor $(1 + \beta A)$, as detailed below. A summary of the gain, feedback factor, and gain with feedback of Fig. 18.2 is provided for reference in Table 18.1.

TABLE 18.1 Summary of Gain, Feedback, and Gain with Feedback from Fig. 18.2

		Voltage-Series	Voltage-Shunt	Current-Series	Current-Shunt
Gain without feedback	A	$\frac{V_o}{V_i}$	$\frac{V_o}{I_i}$	$\frac{I_o}{V_i}$	$\frac{I_o}{I_i}$
Feedback	β	$\frac{V_f}{V_o}$	$\frac{I_f}{V_o}$	$\frac{V_f}{I_o}$	$\frac{I_f}{I_o}$
Gain with feedback	A_f	$\frac{V_o}{V_s}$	$\frac{V_o}{I_s}$	$\frac{I_o}{V_s}$	$\frac{I_o}{I_s}$

VOLTAGE-SERIES FEEDBACK

Figure 18.2a shows the voltage-series feedback connection with a part of the output voltage fed back in series with the input signal, resulting in an overall gain reduction. If there is no feedback ($V_f = 0$), the voltage gain of the amplifier stage is

$$A = \frac{V_o}{V_s} = \frac{V_o}{V_i} \quad (18.1)$$

If a feedback signal, V_f , is connected in series with the input, then

$$V_i = V_s - V_f$$

Since $V_o = AV_i = A(V_s - V_f) = AV_s - AV_f = AV_s - A(\beta V_o)$

then $(1 + \beta A)V_o = AV_s$

so that the overall voltage gain *with* feedback is

$$A_f = \frac{V_o}{V_s} = \frac{A}{1 + \beta A} \quad (18.2)$$

Equation (18.2) shows that the gain *with* feedback is the amplifier gain reduced by the factor $(1 + \beta A)$. This factor will be seen also to affect input and output impedance among other circuit features.

VOLTAGE-SHUNT FEEDBACK

The gain with feedback for the network of Fig. 18.2b is

$$A_f = \frac{V_o}{I_s} = \frac{A I_i}{I_i + I_f} = \frac{A I_i}{I_i + \beta V_o} = \frac{A I_i}{I_i + \beta A I_i}$$

$$A_f = \frac{A}{1 + \beta A} \quad (18.3)$$

14.3 OP-AMP BASICS

An operational amplifier is a very high gain amplifier having very high input impedance (typically a few megohms) and low output impedance (less than 100 Ω). The basic circuit is made using a difference amplifier having two inputs (plus and minus) and at least one output. Figure 14.10 shows a basic op-amp unit. As discussed ear-

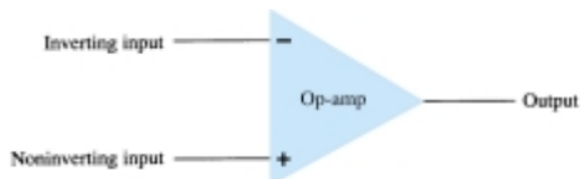


Figure 14.10 Basic op-amp.

lier, the plus (+) input produces an output that is in phase with the signal applied, while an input to the minus (−) input results in an opposite polarity output. The ac equivalent circuit of the op-amp is shown in Fig. 14.11a. As shown, the input signal applied between input terminals sees an input impedance, R_i , typically very high. The output voltage is shown to be the amplifier gain times the input signal taken through an output impedance, R_o , which is typically very low. An ideal op-amp circuit, as shown in Fig. 14.11b, would have infinite input impedance, zero output impedance, and an infinite voltage gain.

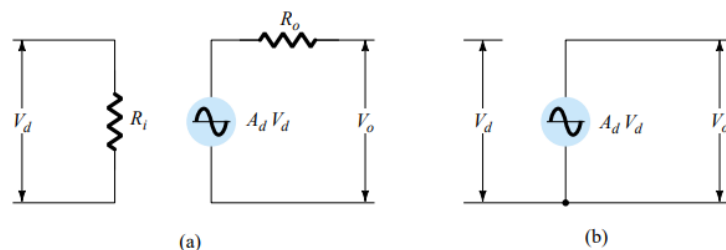


Figure 14.11 Ac equivalent of op-amp circuit: (a) practical; (b) ideal.

Inverting Amplifier

The most widely used constant-gain amplifier circuit is the inverting amplifier, as shown in Fig. 14.15. The output is obtained by multiplying the input by a fixed or constant gain, set by the input resistor (R_i) and feedback resistor (R_f)—this output also being inverted from the input. Using Eq. (14.8) we can write

$$V_o = -\frac{R_f}{R_i} V_i$$

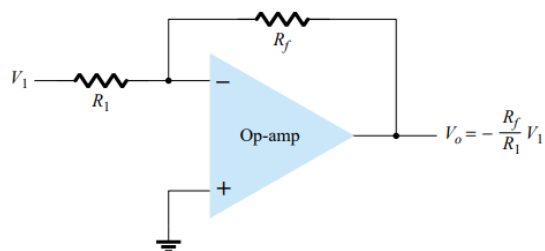


Figure 14.15 Inverting constant-gain multiplier.

Noninverting Amplifier

The connection of Fig. 14.16a shows an op-amp circuit that works as a noninverting amplifier or constant-gain multiplier. It should be noted that the inverting amplifier connection is more widely used because it has better frequency stability (discussed later). To determine the voltage gain of the circuit, we can use the equivalent representation shown in Fig. 14.16b. Note that the voltage across R_1 is V_1 since $V_i \approx 0$ V. This must be equal to the output voltage, through a voltage divider of R_1 and R_f , so that

$$V_1 = \frac{R_1}{R_1 + R_f} V_o$$

which results in

$$\frac{V_o}{V_1} = \frac{R_1 + R_f}{R_1} = 1 + \frac{R_f}{R_1} \quad (14.9)$$

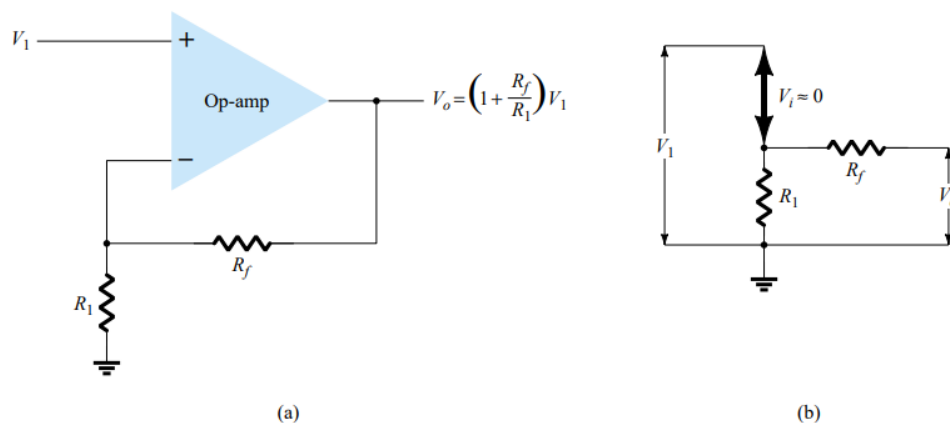


Figure 14.16 Noninverting constant-gain multiplier.

Unity Follower

The unity-follower circuit, as shown in Fig. 14.17a, provides a gain of unity (1) with no polarity or phase reversal. From the equivalent circuit (see Fig. 14.17b) it is clear that

$$V_o = V_1 \quad (14.10)$$

and that the output is the same polarity and magnitude as the input. The circuit operates like an emitter- or source-follower circuit except that the gain is exactly unity.

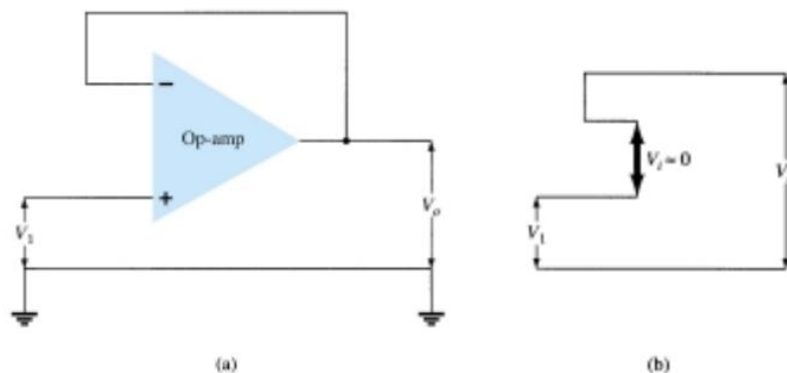


Figure 14.17 (a) Unity follower; (b) virtual-ground equivalent circuit.

Summing Amplifier

Probably the most used of the op-amp circuits is the summing amplifier circuit shown in Fig. 14.18a. The circuit shows a three-input summing amplifier circuit, which provides a means of algebraically summing (adding) three voltages, each multiplied by a constant-gain factor. Using the equivalent representation shown in Fig. 14.18b, the output voltage can be expressed in terms of the inputs as

$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right) \quad (14.11)$$

In other words, each input adds a voltage to the output multiplied by its separate constant-gain multiplier. If more inputs are used, they each add an additional component to the output.

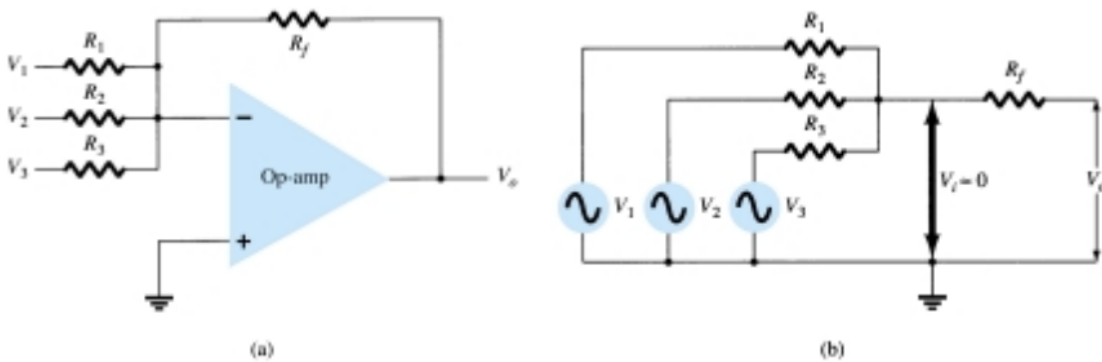


Figure 14.18 (a) Summing amplifier; (b) virtual-ground equivalent circuit.

Integrator

So far, the input and feedback components have been resistors. If the feedback component used is a capacitor, as shown in Fig. 14.19a, the resulting connection is called an *integrator*. The virtual-ground equivalent circuit (Fig. 14.19b) shows that an expression for the voltage between input and output can be derived in terms of the current I , from input to output. Recall that virtual ground means that we can consider the voltage at the junction of R and X_C to be ground (since $V_i \approx 0$ V) but that no current goes into ground at that point. The capacitive impedance can be expressed as

$$X_C = \frac{1}{j\omega C} = \frac{1}{sC}$$

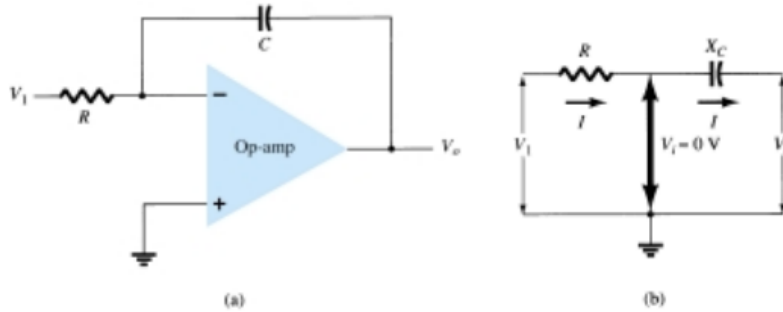


Figure 14.19 Integrator.

where $s = j\omega$ is in the Laplace notation. * Solving for V_o/V_i yields

$$I = \frac{V_i}{R} = -\frac{V_o}{X_C} = \frac{-V_o}{1/sC} = -sCV_o$$

$$\frac{V_o}{V_i} = \frac{-1}{sCR} \quad (14.12)$$

The expression above can be rewritten in the time domain as

$$v_o(t) = -\frac{1}{RC} \int v_i(t) dt \quad (14.13)$$

Differentiator

A differentiator circuit is shown in Fig. 14.22. While not as useful as the circuit forms covered above, the differentiator does provide a useful operation, the resulting relation for the circuit being

$$v_o(t) = -RC \frac{dv_i(t)}{dt} \quad (14.15)$$

where the scale factor is $-RC$.

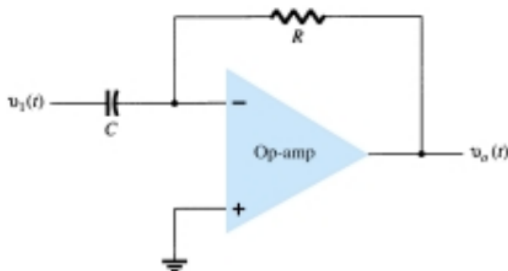


Figure 14.22 Differentiator circuit.